Bitter Pills to Swallow: The Enforcement Costs of Health Litigation

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Abstract

Public health procurement is shaped not only by administrative decisions but also by judicial interventions that compel urgent purchases. Using bid-level data on pharmaceutical procurement in São Paulo, Brazil (2009–2019), with item, time, and buyer fixed effects, we estimate that urgent procurement raises negotiated prices by 3.1–5.4% and reduces bidder participation by 4.1–5.5%. Isolating the sanction channel—comparing litigated and administrative urgent purchases that share identical planning constraints but differ in penalty exposure—we find judicial pressure alone raises prices by 23–30%, driven by higher reference price ceilings and smaller order quantities. Administrative purchases, by contrast, show lower reference prices than ordinary ones, confirming their intermediate position. These findings highlight the economic burden of judicial enforcement on public budgets.

Keywords: public procurement, health litigation, enforcement costs, judicial mandates, Brazil

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1. Introduction

When courts order governments to purchase specific goods, they override standard procurement procedures and compress the time available for planning and competitive bidding. This paper estimates the fiscal costs of such judicial enforcement in the context of health litigation in Brazil, where courts routinely compel state agencies to procure prescription drugs under tight deadlines.

The interaction between the judiciary and public administration has long attracted scholarly attention. Courts and regulatory agencies can be viewed as alternative instruments of social control, each with distinct comparative advantages (Glaeser and Shleifer, 2003). While courts protect individual rights, their interventions in procurement can be costly: inefficient judicial processes may induce public buyers to accept suboptimal outcomes (Coviello et al., 2018). In health care, judicial mandates represent a particularly consequential form of intervention, as they directly determine how—and at what price—the government acquires essential medicines (Wang, 2015; Ferraz, 2009).

Brazil provides an ideal setting to study these costs. The 1988 Constitution enshrines health as a "right of all and a duty of the state," and courts have adopted an expansive interpretation under which virtually any medication can be obtained through litigation, regardless of cost or inclusion on the formulary of the Unified Health System (SUS). Court orders are enforced as preliminary injunctions requiring purchase within days, compared to the weeks or months available for ordinary procurement. The volume of health litigation has grown dramatically: first-instance health claims reached approximately 96,000 nationally in 2017, a 130% increase over 2008 (CNJ/INSPER, 2019). In São Paulo state alone, health lawsuits grew by 913% between 2008 and 2017.

We construct a unique bid-level dataset covering all pharmaceutical procurement transactions by the São Paulo State Department of Health (SES/SP) from January 2009 through December 2019, conducted through the state's electronic procurement platform (BEC). The dataset comprises over 479,000 purchase-offer-item (POI) observations for medical supplies (BEC Group 65), encompassing ordinary, administrative, and litigated purchases. We classify purchase types using a regular expression algorithm applied to public tender notices, cross-referenced with administrative records from the S-CODES health litigation database.

Our empirical strategy exploits the quasi-experimental variation in procurement conditions generated by court orders and administrative requests. These exogenous shocks partition purchases into two regimes: *ordinary* (standard planning timeline) and *urgent* (compressed timeline due to judicial or administrative mandate). Within urgent purchases, we further distinguish *administrative* requests—which carry no penalty for procurement failure—from *litigated* purchases, where officials face severe sanctions including fines, personal liability, and fund seizure. This distinction allows us to isolate what we term the "under the gun" effect: the additional cost attributable to the threat of judicial punishment.

We estimate high-dimensional fixed effects regressions with item, year (or year-month), and public buyer unit (PBU) fixed effects, clustering standard errors at the PBU level. Our preferred specification (item + year + PBU FE) compares the same product purchased by the same buyer across ordinary and urgent procurement episodes, absorbing time-invariant item heterogeneity, common time trends, and buyer-specific factors.

The main findings are as follows. First, urgent purchases carry higher reference prices—the maximum price the government is willing to pay—by 2.7% in our preferred specification. Second, negotiated prices are 5.4% higher for urgent purchases (total effect), with a direct effect of 3.1% after controlling for quantity differences. Third, urgent tenders attract 5.5% fewer bidding firms. Fourth, the probability of tender success is 2.1 percentage points *higher* for urgent purchases, consistent with procurement officials accepting higher prices to ensure compliance rather than risk tender failure.

We further isolate the effect of court orders by comparing litigated purchases directly to ordinary ones, excluding administrative purchases. This yields larger estimates across all outcomes: reference prices are 7.6% higher, negotiated prices 9.1% higher, and bidder participation 7.9% lower for litigated purchases. The entire negotiated price premium operates through the quantity channel: controlling for quantity, the direct effect is economically zero. Complementing this analysis, we show that administrative purchases—which share urgency constraints but lack judicial sanctions—actually carry *lower* reference prices than ordinary purchases, confirming the intermediate position of administrative procurement and isolating the sanction channel as the primary cost driver.

The "under the gun" analysis reveals that the sanction channel is quantitatively important. Comparing litigated and administrative urgent purchases for the same items, we find that litigated tenders carry prices 23.2–30.2% higher in our preferred specifications, a difference attributable to the threat of judicial punishment. Extending this analysis to all outcomes, we show that the sanction channel operates primarily through reference prices—which are 27% higher for litigated than administrative purchases—and through quantities, with litigated purchases involving substantially smaller orders that forgo bulk discounts.

Our heterogeneity analysis shows that the enforcement costs vary across several dimensions. The price premium of urgent purchases is concentrated among items in the specialized SUS component (non-basic medicines), in the later period (2014–2019), and in markets with lower competition. These patterns are consistent with a model in which judicial pressure interacts with market structure to amplify procurement inefficiency.

This paper makes three contributions to the literature. First, we provide the first estimates of the causal effect of judicial enforcement on public procurement outcomes using comprehensive administrative data and high-dimensional fixed effects, contributing to the literature on government spending efficiency (Bandiera et al., 2009; Best et al., 2023; Decarolis et al., 2020). Second, we identify and quantify the "under the gun" effect—the cost premium attributable to the threat of judicial sanctions—by exploiting the institutional distinction between administrative and litigated urgent purchases. This contributes to the growing literature on how oversight and accountability mechanisms can generate unintended costs (Rasul and Rogger, 2018; Williams, 2021; Duflo et al., 2012). Third, we document how judicial mandates distort the planning phase of procurement, adding to the evidence on the importance of ex ante design in auction performance (Coviello et al., 2018; Szücs, 2023; Baltrunaite et al., 2021).

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the institutional background. Section 4 presents the data and sample. Section 5 details the empirical strategy. Section 6 presents the results. Section 7 concludes.

2. Related Literature

This paper contributes to several strands of the economics literature.

Government spending efficiency and procurement.. A foundational distinction in the study of public spending separates *active waste* (corruption) from *passive waste* (inefficiency), as formalized by Bandiera et al. (2009), who estimate that passive waste accounts for the majority of excess spending in Italian public procurement. Subsequent work has identified specific channels of passive waste, including bureaucratic capacity (Best et al., 2023), management practices (Rasul and Rogger, 2018; Bloom et al., 2015), and the design of procurement mechanisms (Lewis-Faupel et al., 2016; Szücs, 2023). Our paper identifies a previously unstudied source of passive waste: the compression of planning time induced by judicial mandates. We show that this channel operates through both the reference price setting process (the internal phase) and the competitive bidding process (the external phase).

Procurement design and competition.. A growing literature documents how procurement design affects competition and prices. Coviello et al. (2018) show that court-imposed delays reduce bidder participation and increase procurement costs in Italian public works. Decarolis et al. (2020) demonstrate that buyer competence is a key determinant of procurement outcomes. Baltrunaite et al. (2021) find that reducing discretion in procurement can improve outcomes when corruption is prevalent but may harm outcomes otherwise. Our setting differs in that judicial intervention compresses rather than extends the procurement timeline, and the primary mechanism operates through planning disruption rather than direct manipulation of award criteria.

Accountability, oversight, and unintended consequences.. A related literature examines how accountability mechanisms intended to reduce waste can themselves generate costs. Rasul and Rogger (2018) show that management practices in the Nigerian civil service affect project completion rates, while Williams (2021) finds that management quality explains substantial variation in public service delivery. Duflo et al. (2012) argue that the design of incentives for public officials must balance accountability with operational flexibility. Our "under the gun" effect is a specific instance of this tension: judicial sanctions intended to ensure compliance with health mandates inadvertently raise procurement costs by distorting the bargaining environment.

Health litigation and right-to-health enforcement.. The legal and public health literatures have documented the growth of health litigation in developing countries, particularly in Latin America (Wang, 2015; Ferraz, 2009; Biehl et al., 2009). CNJ/INSPER (2019) provide comprehensive data on the judicialization of health in Brazil. However, the existing literature has focused primarily on the legal and public health dimensions, with limited attention to the fiscal costs transmitted through the procurement channel. Our paper fills this gap by providing granular estimates of how judicial enforcement affects specific procurement outcomes—prices, quantities, competition, and tender success.

Auction theory under urgency.. From a theoretical perspective, our findings connect to the auction theory literature on how the number of bidders and time pressure affect prices. Bulow and Klemperer (1996) show that attracting an additional bidder can be more valuable than optimizing the auction mechanism. When judicial mandates reduce bidder participation, they directly undermine the competitive benefits of procurement auctions. Moreover, the public information that a tender is court-mandated creates an asymmetric bargaining environment that further erodes the government's position (Krishna, 2009).

3. Institutional Background

This section describes the institutional framework governing health litigation and pharmaceutical procurement in Brazil, with a focus on the state of São Paulo.

3.1. Health as a Constitutional Right and the Rise of Litigation

The 1988 Brazilian Constitution created the Unified Health System (SUS), a universal public health system jointly funded and administered by federal, state, and municipal governments. The SUS maintains a formulary of approved medications and procedures, periodically updated to reflect medical advances subject to cost-effectiveness criteria. This formulary represents the society's chosen balance between universal coverage and fiscal sustainability (Castro et al., 2019; Soares, 2019).

However, Brazilian courts have adopted a strict literal interpretation of Article 196 of the Constitution, which states that "health is a right of all and a duty of the state." Under this interpretation, any individual can obtain virtually any medication through litigation, irrespective of cost or inclusion on the SUS formulary. The range extends from acetylsalicylic acid to galsulfase (for mucopolysaccharidosis type VI, with annual treatment costs of approximately US$400,000).

Several factors make health litigation particularly prevalent. Access to the legal system is inexpensive—requiring only a prescription and a public defender—and success rates are exceptionally high: approximately 85% of first-instance claims in São Paulo are granted (CNJ/INSPER, 2019), with even higher rates on appeal. These conditions create strong incentives for individuals to seek medicines through the courts rather than through standard SUS channels.

The growth in health litigation has been dramatic. Nationally, first-instance health court orders reached approximately 96,000 in 2017, representing a 130% increase over 2008 (CNJ/INSPER, 2019). In São Paulo, the increase was 913%, from 2,317 to 23,465 annual lawsuits between 2008 and 2017. Court orders span a wide geographic distribution across the state's municipalities (Figure 1) and encompass approximately 2,760 distinct items ordered at least once per year.

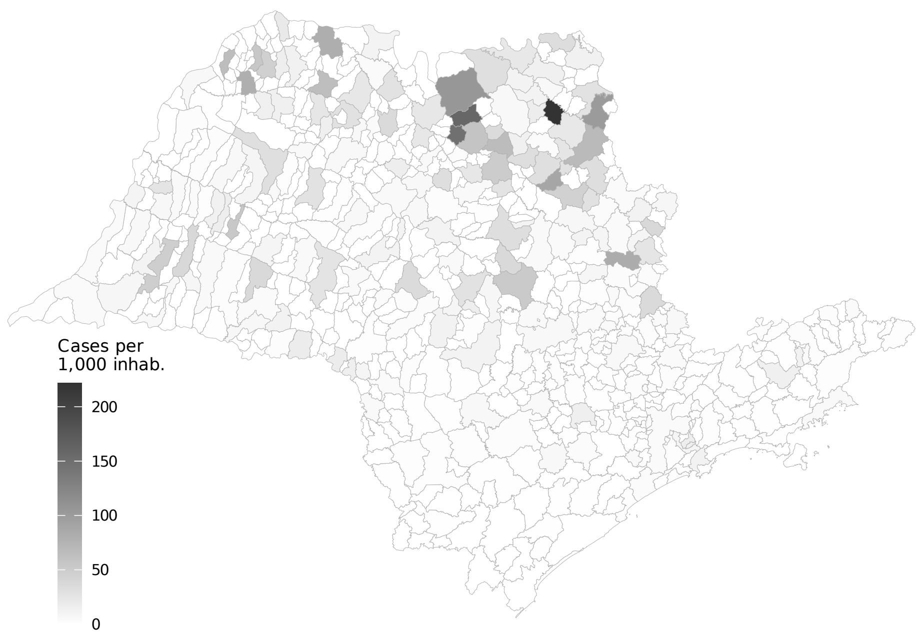

Figure 1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)
Source: Authors' elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates (SIDRA table 6579, 2009–2019 average).

The fiscal impact is substantial. São Paulo's annual public health budget was approximately US$5.9 billion in 2018 (about 10% of the state's total budget). Public spending on litigated health items alone amounted to nearly 5% of this budget, or approximately US$300 million annually.

3.2. Public Procurement under Judicial Pressure

The São Paulo State Department of Health (SES/SP) manages health policy implementation through 99 decentralized public buyer units (PBUs) distributed across the state (Figure 2).

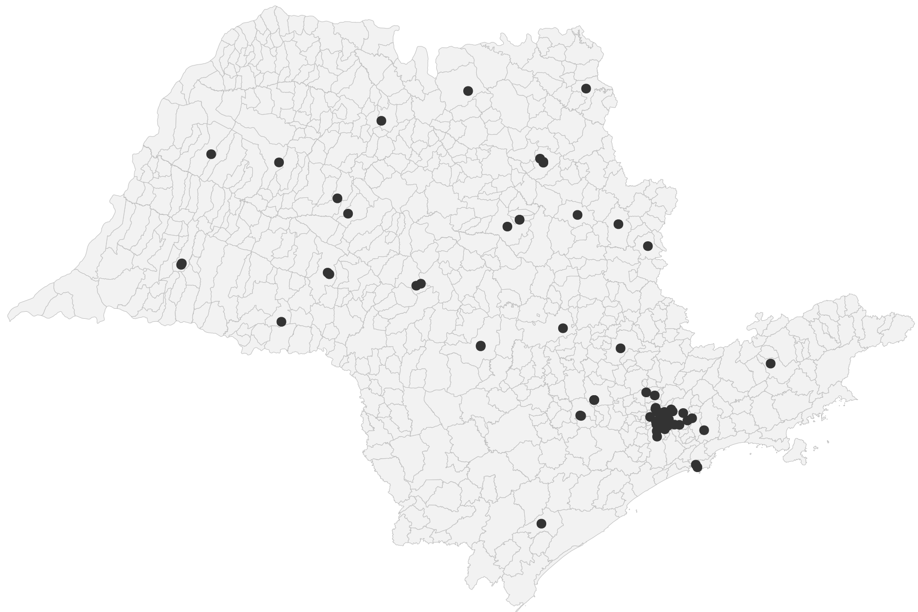

Figure 2: Public Buyer Units (SES/SP)
Source: Authors' elaboration based on SES/SP administrative records.

Federal Law 8,666/1993 establishes the general framework for public procurement, requiring that all government purchases follow a formal tendering process organized in three sequential phases:

1. **Internal phase (planning):** PBUs identify needs, specify items, set quantities and reference prices, choose the auction format, and publish a public notice.
2. **External phase (bidding):** Firms compete through a chosen competitive procedure—either sealed-bid (*convite*, up to R$176,000) or multi-round descending auction (*pregão*, no value limit).
3. **Delivery phase:** The winning firm delivers the procured items.

For ordinary purchases, the internal phase typically spans 30 to 180 days, allowing PBUs to aggregate demand, research market prices, and set competitive reference prices. Court orders fundamentally disrupt this process. Almost all court decisions (99.94%) are enforced as preliminary injunctions with delivery deadlines of 1 to 10 days, compressing the internal phase to roughly one-third of the ordinary timeline.

This compression affects procurement in several ways. First, PBUs lose the ability to aggregate demand across units and over time, reducing their bargaining leverage from bulk purchasing. Second, reference prices must be set quickly with limited market research, often resulting in higher reservation prices. Third, approximately 75% of litigated items are not on the SUS formulary, meaning PBUs have less experience procuring them. Fourth, roughly 65% of court orders involve "packages" combining SUS and non-SUS items, further complicating procurement planning.

3.3. The Sanction Channel: Administrative vs. Litigated Purchases

A critical institutional feature for our identification strategy is the distinction between *litigated* and *administrative* urgent purchases. Both types face identical planning constraints—compressed timelines, small quantities, and diverted budget resources. However, they differ sharply in the consequences of procurement failure.

For litigated purchases, failure to comply with a court order exposes public officials to severe sanctions: substantial fines (sometimes disproportionate to the purchase value), administrative and criminal liability, and seizure or blocking of public funds. These penalties are public information, as the court order must be referenced in the tender notice.

Since 2009, the SES/SP has attempted to mitigate these costs through administrative requests, a mechanism allowing the government to negotiate supply directly with individuals before litigation occurs. Administrative requests undergo evaluation by a scientific committee using evidence-based medicine criteria. Crucially, failure to complete an administrative purchase carries no penalties for public officials. Between 2009 and 2019, administrative requests generated approximately 9,700 purchases, or about 6.5% of the total from court orders.

This institutional distinction provides a natural comparison group for isolating the "under the gun" effect—the additional procurement cost attributable to the threat of judicial sanctions—since administrative and litigated purchases share all planning and execution constraints but differ only in the penalty regime.

	Ordinary	Administrative	Litigated
Source of funds	Dedicated budget	Diverted budget	Diverted budget
Quantity	Large	Small	Small
Delivery time	Long	Short	Short
Threat of punishment	None	None	Potential punishment

Figure 3: Types of Purchases
Source: Authors' elaboration.

3.4. The Bidding Process

Both ordinary and urgent purchases are conducted through the *Bolsa Eletrônica de Compras* (BEC), São Paulo state's electronic procurement platform, mandatory for all common goods and services since 2007. The BEC handles substantial volumes: approximately US$1.7 billion in SES/SP transactions in 2018, and over R$7.4 billion since 2009.

Two competitive procedures are used. In sealed-bid tendering (*convite*), firms submit proposals by a deadline; the lowest qualifying bid wins. In multi-round descending auctions (*pregão*), an initial sealed-bid phase is followed by a 20-minute reverse auction where firms observe the current lowest bid and can submit lower bids. If a bid is placed in the final four minutes, the auction extends by four additional minutes. After the auction, a negotiation phase allows further price reductions.

The key implication for our analysis is that all purchase types use the same electronic platform and competitive procedures. The variation we exploit arises exclusively from differences in planning conditions and penalty regimes, not from differences in the auction mechanism itself.

4. Data and Sample

4.1. Data Sources

We combine three administrative datasets covering January 2009 through December 2019.

Public procurement data.. Our primary dataset consists of bid-level records from the BEC electronic procurement platform for all SES/SP purchases of medical supplies classified under BEC

Group 65 ("Medical, dental, and hospital equipment and supplies"). Each observation corresponds to a purchase-offer-item (POI)—the combination of a purchase order and a specific item—and includes information on: (i) the internal phase (quantities demanded, reference prices, PBU identifier, auction format); (ii) the external phase (bid prices from all participants, number of bidding firms, tender outcome); and (iii) the winning bid (final negotiated price, firm identifier).

Purchase type classification.. We classify each POI as ordinary, administrative, or litigated using a regular expression (REGEX) algorithm applied to the "Object of the Contract" field of public tender notices. This text field describes the purchase purpose and, by law, must reference any underlying court order or administrative request.

Health litigation records.. Data on individual health claims come from S-CODES, the SES/SP administrative database of all health-related lawsuits against the state of São Paulo. We derive several variables from this source, including SUS formulary status, whether the claim involves a drug package, and whether the order was enforced through a preliminary injunction.

4.2. Sample Construction

The raw dataset comprises 479,330 POI observations across 32,532 distinct items, 94,182 purchase orders, and 100 PBUs. Table 1 reports descriptive statistics by purchase type.

The analysis sample is restricted to items for which at least one ordinary and at least one litigated purchase is observed during the sample period, ensuring within-item comparability across procurement regimes. This restriction yields a sample of approximately 226,000 observations covering 3,856 items. For regressions using bid prices and quantities, we further restrict to winning bids (approximately 197,000 observations).

Several features of the data are noteworthy. In levels, ordinary purchases have substantially higher mean reference prices (R$909) compared to administrative (R$226) and litigated (R$370) purchases, reflecting the mix of items across procurement types. However, the composition effect is absorbed by our item fixed effects. In logs—which form the basis of our regressions—litigated purchases show higher mean prices (1.56 for reference price, 1.02 for negotiated price) than ordinary purchases (0.93 and 0.32, respectively), consistent with a price premium for urgent procurement conditional on the item.

Table 1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O–L)	(A–L)
Panel A: Levels								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78*** [0.000]	-144.01*** [0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26*** [0.000]	-112.22*** [0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96*** [0.000]	107,499.81*** [0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97*** [0.000]	0.23*** [0.000]
Panel B: Log Transformations								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632*** [0.000]	-0.648*** [0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705*** [0.000]	-0.544*** [0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860*** [0.000]	1.385*** [0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319*** [0.000]	0.081*** [0.000]
Panel C: Tender Characteristics								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050*** [0.000]	-0.040*** [0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch t-tests; p-values in brackets. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Quantities are markedly lower for litigated purchases (mean of 9,646 units vs. 31,487 for ordinary), reflecting the inability to aggregate demand under compressed timelines. The number of bidding firms is also lower for urgent purchases (2.2 for litigated vs. 3.2 for ordinary), suggesting reduced competition in the external phase.

For the "under the gun" analysis, we construct a subsample of urgent purchases (administrative and litigated) for items with both types present, yielding approximately 57,000 observations. Table 2 presents balance statistics for this subsample.

Table 2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	p-value
	Mean	(SD)	Mean	(SD)	(A–L)	
Panel A: Procurement Outcomes						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
Panel B: Market Structure						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
Panel C: Purchase Characteristics						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. p-values from Welch t-tests in brackets. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

4.3. Variable Definitions

Our key variables are defined as follows:

- **Urgent:** indicator equal to 1 for administrative or litigated purchases (purchase type $\in \{1,2\}$), 0 for ordinary purchases.
- **Administrative:** indicator equal to 1 for administrative purchases, 0 for litigated. Used in the "under the gun" subsample analysis.
- **Reference price:** the maximum price specified in the tender notice. We use the log transformation.
- **Negotiated price:** the final price of the winning bid. Log transformation.
- **Quantity:** the quantity demanded in the tender notice. Log transformation.
- **Number of firms:** the count of distinct firms submitting bids. Log transformation.
- **Tender success:** indicator equal to 1 if the tender resulted in a successful purchase.

All continuous variables are winsorized at the 1st and 99th percentiles to limit the influence of outliers. Robustness checks with no winsorization and 5%/95% winsorization are reported in the Appendix.

5. Empirical Strategy

Our empirical analysis proceeds in two stages. First, we estimate the overall enforcement costs of urgent procurement by comparing ordinary and urgent purchases. Second, we isolate the "under the gun" effect by comparing litigated and administrative urgent purchases.

5.1. Identification

The key identification assumption is that, conditional on item, time, and buyer fixed effects, the assignment of purchases to the urgent category is exogenous to unobserved determinants of procurement outcomes. Several institutional features support this assumption.

First, judicial orders and administrative requests originate from individuals' health needs and legal claims, which are unrelated to the procurement process itself. The principle of impersonality in Brazilian public administration ensures that tender outcomes depend on purchase characteristics—item specifications, market conditions, and planning parameters—not on the identity or circumstances of the requesting individual.

Second, the timing and content of court orders are unpredictable from the PBU's perspective. PBUs cannot anticipate exactly which items, in what quantities, and when they will be compelled to purchase. Only 4% of judicial claims arise from stock shortages or service failures, suggesting that litigation is poorly correlated with PBU-level mismanagement or unobserved quality.

Third, the key source of variation—the urgency regime—is determined by external judicial or administrative decisions, not by PBU choice. While PBUs might in principle manipulate item classification, the legal requirement to reference court orders in tender notices and our REGEX-based classification using official texts mitigate this concern.

5.2. Enforcement Costs: Ordinary vs. Urgent

We estimate the following baseline specification:

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{1}$$

where $y_{i,g,t}$ is the outcome (log reference price, log negotiated price, log quantity, log number of firms, or tender success) for purchase order i of item g at time t; $\text{Urgent}_{i,g,t}$ equals 1 for administrative or litigated purchases and 0 for ordinary purchases; γ_g are item fixed effects; λ_t are time fixed effects (year or year-month); and δ_b are PBU fixed effects. Standard errors are clustered at the PBU level.

We report four specifications with progressively richer fixed effects:

(1) Item FE only;

(2) Item + Year FE;

(3) Item + Year + PBU FE (preferred);

(4) Item + Year-Month + PBU FE.

Specification (3) is our preferred specification because it absorbs time-invariant item characteristics, common annual trends, and buyer-specific heterogeneity, while retaining sufficient within-cell variation for precise estimation.

For negotiated prices and number of firms, we additionally estimate a "direct effect" specification that controls for log quantity:

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \alpha \cdot \ln Q_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}. \tag{2}$$

This specification captures the direct effect of urgency on prices and competition, net of the quantity channel (since urgent purchases typically involve smaller quantities, which may independently affect prices through reduced bulk discounts).

5.3. The "Under the Gun" Effect: Litigated vs. Administrative

To isolate the effect of judicial sanctions on procurement costs, we restrict the sample to urgent purchases only (both administrative and litigated) for items with at least one purchase of each type, and estimate:

$$\ln(\text{Price}_{i,g,t}) = \beta \cdot \text{Admin}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{3}$$

where Admin equals 1 for administrative purchases and 0 for litigated purchases. The coefficient β captures the price differential between administrative and litigated purchases, holding constant the item, time period, and buying unit. Since both types face identical planning constraints, β isolates the cost attributable to the sanction regime: the threat of judicial punishment for procurement failure.

6. Results

6.1. Overview

We organize the results around four comparisons: (i) urgent vs. ordinary purchases, (ii) the "under the gun" effect (litigated vs. administrative), (iii) litigated vs. ordinary, and (iv) administrative vs. ordinary. Figure 4 provides a geographic overview of the three purchase types across São Paulo municipalities.

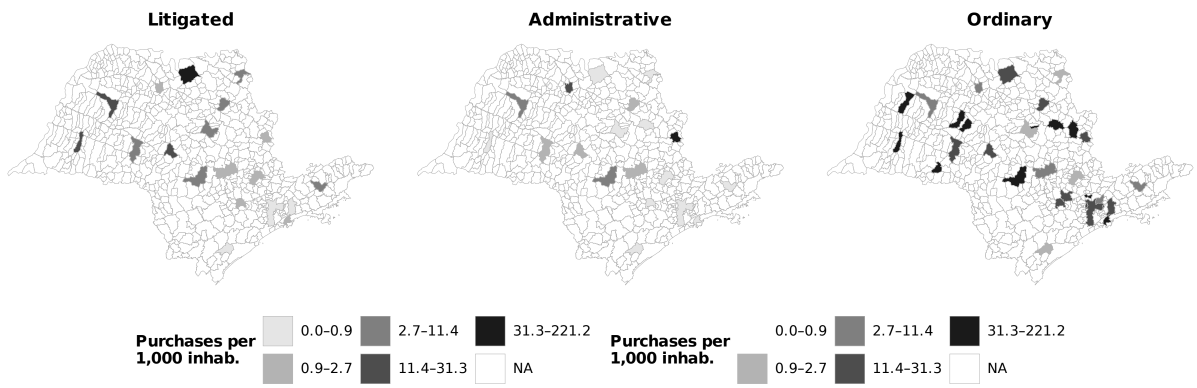

Figure 4: Geographic Distribution of Purchases per 1,000 Inhabitants: Litigated, Administrative, and Ordinary (São Paulo, 2009–2019). Quintile breaks; SIRGAS 2000 projection.

Source: Authors' elaboration based on BEC procurement data and IBGE population estimates.

6.2. Urgent vs. Ordinary Purchases

We begin with the pooled comparison of urgent (administrative + litigated) vs. ordinary purchases. Table A.1 (Appendix) reports reference price estimates: urgent purchases carry a 2.7% premium in the preferred specification ($p < 0.10$), with the raw premium attenuating substantially once PBU fixed effects are included. Table A.2 (Appendix) shows imprecisely estimated negative effects on quantities (-5.6% in the preferred specification).

Table 3 presents the core results on negotiated prices.

Table 3: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel B: Direct Effect				
Urgent Purchase	0.075	0.060	0.030*	0.030*
	(0.045)	(0.042)	(0.017)	(0.017)
Log Quantity	-0.321***	-0.325***	-0.392***	-0.393***
	(0.026)	(0.026)	(0.029)	(0.029)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883
Within R^2 (A)	0.003	0.003	0.000	0.000
Within R^2 (B)	0.315	0.327	0.358	0.359

Notes: Dependent variable: log negotiated price. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Total effect (Panel A).. Negotiated prices are significantly higher for urgent purchases across all specifications. The total effect ranges from 17.2% with item FE only to 5.4–5.8% in our preferred specifications (columns 3–4). The attenuation reflects the importance of buyer-level heterogeneity.

Direct effect (Panel B).. Controlling for log quantity reduces the coefficient by roughly half: from 0.053 to 0.030 (column 3). The direct effect of 3.1% ($p < 0.10$) represents the price premium net of the quantity channel. The large negative coefficient on log quantity (-0.392) confirms the importance of bulk discounts.

Bidder participation falls by 5.5% for urgent purchases (4.1% direct effect), and urgent tenders are 2.1 pp more likely to succeed (Tables A.3–A.4, Appendix). The higher success rate despite worse terms is consistent with procurement officials prioritizing compliance over cost-efficiency. Figure 5 provides a visual summary of these findings.

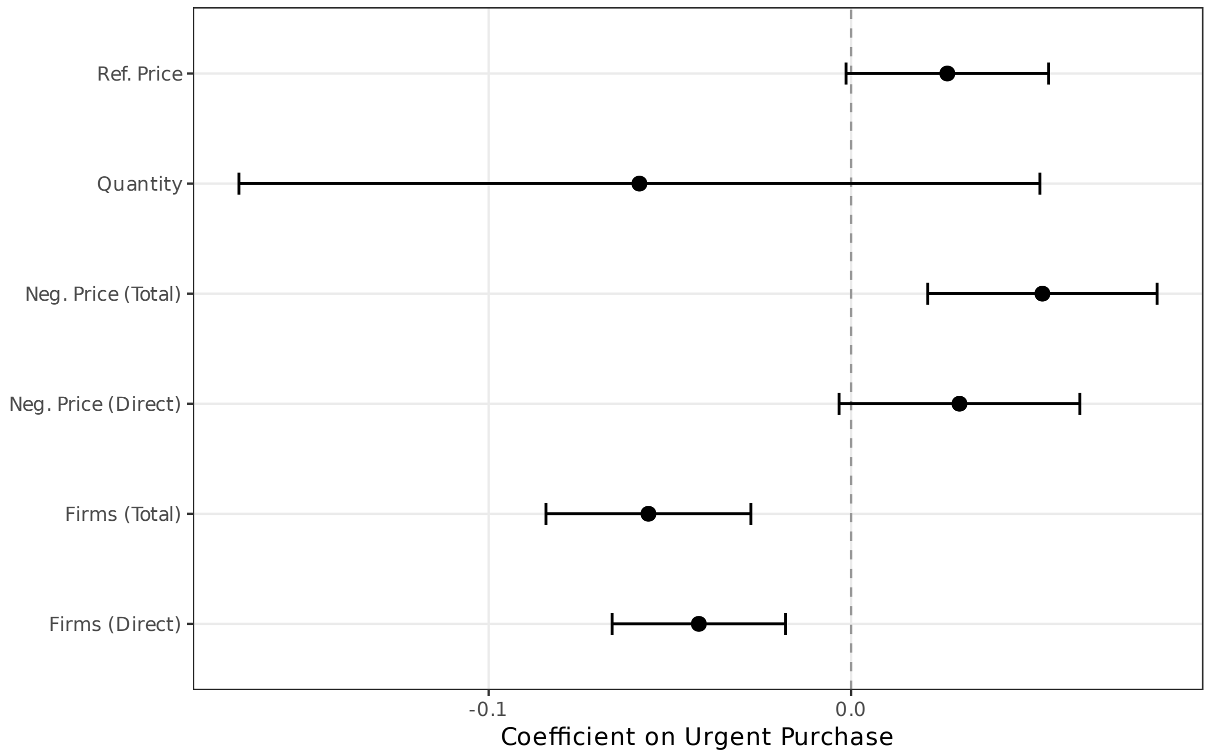

Figure 5: Urgent vs. Ordinary: Coefficient Estimates with 95% Confidence Intervals (Preferred Specification)

6.3. The "Under the Gun" Effect

The "under the gun" analysis restricts the sample to urgent purchases only and estimates the effect of the administrative indicator. Since both types share identical planning constraints but differ in penalty exposure, a negative coefficient on the administrative indicator implies that litigated purchases are more expensive.

Table 4 presents the core test for the sanction channel.

Total effect (Panel A).. Administrative purchases have significantly lower negotiated prices than litigated ones. In our preferred specification (column 3), the coefficient is -0.262 ($p < 0.05$), implying that litigated purchases are 30.0% more expensive than comparable administrative purchases.

Direct effect (Panel B).. Controlling for log quantity, the coefficient becomes smaller and insignificant, suggesting that the price premium operates substantially through the quantity channel: litigated purchases involve smaller quantities, which translate into higher per-unit prices.

Table 4: Under the Gun: Administrative vs Litigated

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Administrative (vs Litigated)	-0.209***	-0.205**	-0.262**	-0.265***
	(0.074)	(0.074)	(0.096)	(0.094)
Panel B: Direct Effect				
Administrative (vs Litigated)	-0.006	0.016	0.117	0.115
	(0.056)	(0.058)	(0.122)	(0.115)
Log Quantity	-0.284***	-0.292***	-0.341***	-0.344***
	(0.036)	(0.038)	(0.042)	(0.040)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	56,803	56,803	56,803	56,802
Within R^2 (A)	0.011	0.010	0.007	0.007
Within R^2 (B)	0.287	0.292	0.307	0.310

Notes: Dependent variable: log negotiated price. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Extending this analysis to all outcomes (Tables A.5–A.8, Appendix), we find that the sanction channel operates primarily through reference prices—which are 27% higher for litigated than administrative purchases—and through quantities, with litigated purchases involving substantially smaller orders. By contrast, bidder participation and tender success rates do not differ significantly, indicating that the competition channel is driven by urgency per se rather than judicial pressure specifically.

6.4. *Litigated vs. Ordinary Purchases*

To isolate the specific effect of court-mandated procurement, we replicate the main regressions excluding administrative purchases entirely. The results (Tables A.9–A.13, Appendix) yield larger effects across all outcomes compared to the pooled analysis: reference prices are 7.6% higher, quantities 20.7% lower, negotiated prices 9.1% higher, and bidder participation 7.9% lower ($p < 0.01$ for all). The near-zero direct effect on negotiated prices after controlling for quantity confirms the central role of the quantity channel. These magnitudes confirm that the fiscal costs of urgency are concentrated in court-mandated procurement.

6.5. Administrative vs. Ordinary Purchases

As a complementary comparison, we estimate the effect of administrative purchases relative to ordinary ones, excluding litigated purchases from the sample. The results (Tables A.14–A.18, Appendix) reveal a strikingly different pattern from the litigated comparison. Administrative purchases carry *lower* reference prices than ordinary ones (-9.1%, $p < 0.10$) and show weakly lower prices on other dimensions, though most coefficients are imprecisely estimated. This confirms that administrative purchases occupy an intermediate position: they share the planning constraints of urgent procurement but lack the price-inflating effects of judicial sanctions. The contrast with the litigated-only estimates reinforces the conclusion that the "under the gun" effect—not urgency per se—is the primary driver of procurement costs.

6.6. Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions (Tables A.19–A.22, Appendix). The price premium is concentrated among basic SUS items (0.030, $p < 0.10$) rather than specialized items, suggesting that procurement disruption is most consequential where competitive markets normally deliver better prices. The premium is larger in the later period (2014–2019), consistent with the growing volume of litigation. Markets with lower competition and smaller PBUs show somewhat larger premia, though differences are not statistically significant.

7. Conclusion

This paper estimates the fiscal costs of judicial enforcement in public health procurement, exploiting granular bid-level data on pharmaceutical purchases in São Paulo, Brazil. Our findings document that court-mandated urgent procurement significantly distorts procurement outcomes: reference prices rise by 2.7%, negotiated prices increase by 3.1–5.4%, and bidder participation falls by 4.1–5.5%, all within item, year, and buyer comparisons.

The "under the gun" analysis reveals that the sanction channel is particularly consequential. Comparing litigated and administrative urgent purchases—which share identical planning constraints but differ in penalty exposure—we find that judicial sanctions alone account for a 23–30% price premium. This premium is driven by two specific mechanisms: litigated purchases carry

27% higher reference price ceilings, reflecting ex ante planning distortions, and involve substantially smaller quantities that forgo bulk discounts. By contrast, bidder participation and tender success rates do not differ significantly between administrative and litigated urgent purchases, indicating that the competition channel is driven by urgency per se rather than judicial pressure specifically. Comparing litigated purchases directly to ordinary ones—excluding administrative purchases entirely—confirms that court-mandated procurement drives the bulk of the enforcement costs, with effects exceeding those from the pooled urgent analysis across all outcome dimensions. Notably, administrative purchases carry *lower* reference prices than ordinary ones, confirming that urgency without judicial sanctions does not inflate procurement costs—it is the sanction regime, not compressed timelines per se, that drives the fiscal burden.

The finding that urgent tenders are more likely to succeed, despite worse terms, is consistent with the mechanism: procurement officials under judicial pressure prioritize compliance over cost-efficiency, accepting higher prices to avoid the severe consequences of tender failure. This pattern echoes the broader insight from the accountability literature that oversight mechanisms can generate unintended costs when they distort rather than align incentives (Rasul and Rogger, 2018; Duflo et al., 2012).

Our results carry several policy implications. First, the large cost premium attributable to judicial sanctions suggests that revising penalty mechanisms could yield significant fiscal savings. More proportionate sanctions—or institutional arrangements that shield procurement officials from personal liability for good-faith efforts—could preserve accountability while reducing the distortionary "under the gun" effect. Second, enhanced coordination between the judiciary and the executive, such as expanding the administrative request mechanism, could reduce the volume of litigated purchases and the associated costs. Third, the concentration of enforcement costs in markets with less competition suggests that targeted supply-side interventions—such as framework agreements or strategic stockpiling for commonly litigated items—could mitigate the price premium.

More broadly, our findings highlight a tension at the heart of right-to-health enforcement: judicial mandates intended to guarantee individual access to health care impose substantial costs on the public budget, potentially reducing the resources available for the broader health system. The estimated price premiums, applied to the approximately US$300 million spent annually on litigated health items in São Paulo alone, imply tens of millions of dollars in excess costs. Institutional designs that balance individual rights with procurement efficiency—rather than treating them as competing objectives—could improve welfare for both litigants and the broader population.

Several limitations suggest directions for future research. Our sample covers a single state's procurement system, and the generalizability of the magnitudes to other jurisdictions and institutional contexts remains an open question. The "under the gun" identification relies on the assumption that administrative and litigated purchases differ only in the penalty regime; while institutional evidence strongly supports this assumption, unobserved differences in the composition of requests cannot be entirely ruled out. Future work could exploit administrative reforms or policy changes that alter the penalty regime to provide complementary identification. Cross-country comparisons of judicial enforcement in procurement—particularly in other Latin American countries with active health litigation—would also be valuable.

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Appendix

A.1 Additional Main Results

Tables A.1–A.4 report the full set of urgent vs. ordinary estimates for outcomes not shown in the main text.

Table A.1: Reference Prices

	(1)	(2)	(3)	(4)
Urgent Purchase	0.164***	0.156***	0.027*	0.027*
	(0.056)	(0.050)	(0.014)	(0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R^2	0.003	0.003	0.000	0.000

Notes: Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.2: Quantities

	(1)	(2)	(3)	(4)
Urgent Purchase	-0.264	-0.279	-0.058	-0.065
	(0.183)	(0.181)	(0.056)	(0.058)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R^2	0.003	0.003	0.000	0.000

Notes: Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.3: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.014)	(0.015)
Panel B: Direct Effect				
Urgent Purchase	-0.091***	-0.102***	-0.042***	-0.040***
	(0.014)	(0.011)	(0.012)	(0.013)
Log Quantity	0.063***	0.064***	0.065***	0.065***
	(0.004)	(0.004)	(0.003)	(0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,805	81,805	81,801	81,801
Obs. (Panel B)	81,793	81,793	81,789	81,789
Within R^2 (A)	0.004	0.006	0.001	0.001
Within R^2 (B)	0.063	0.069	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.4: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel B: Direct Effect				
Urgent Purchase	0.023**	0.023**	0.021***	0.022***
	(0.009)	(0.010)	(0.006)	(0.006)
Log Quantity	0.003	0.003	0.012***	0.012***
	(0.004)	(0.004)	(0.002)	(0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	226,305	226,305	226,301	226,301
Obs. (Panel B)	226,282	226,282	226,278	226,278
Within R^2 (A)	0.001	0.001	0.000	0.000
Within R^2 (B)	0.001	0.001	0.004	0.004

Notes: Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

A.2 Under the Gun: Extended Outcomes

Tables A.5–A.8 extend the UTG analysis to reference prices, quantities, bidder participation, and tender success. Figure A.1 provides a visual summary.

Table A.5: Under the Gun: Reference Prices

	(1)	(2)	(3)	(4)
Administrative (vs Litigated)	-0.307***	-0.297***	-0.319***	-0.321***
	(0.087)	(0.089)	(0.087)	(0.086)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	56,804	56,804	56,804	56,803
Within R^2	0.024	0.023	0.012	0.012

Notes: Dependent variable: log reference price. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.6: Under the Gun: Quantities

	(1)	(2)	(3)	(4)
Administrative (vs Litigated)	0.712	0.758*	1.115*	1.106*
	(0.434)	(0.426)	(0.618)	(0.588)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	56,804	56,804	56,804	56,803
Within R^2	0.035	0.039	0.048	0.046

Notes: Dependent variable: log quantity. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.7: Under the Gun: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Administrative (vs Litigated)	0.045	0.071	0.102	0.103
	(0.052)	(0.049)	(0.070)	(0.069)
Panel B: Direct Effect				
Administrative (vs Litigated)	-0.028	-0.002	0.031	0.031
	(0.020)	(0.020)	(0.037)	(0.037)
Log Quantity	0.069***	0.066***	0.067***	0.068***
	(0.005)	(0.004)	(0.005)	(0.005)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	15,053	15,053	15,052	15,052
Within R^2 (A)	0.002	0.005	0.004	0.004
Within R^2 (B)	0.069	0.067	0.046	0.047

Notes: Dependent variable: log number of bidding firms. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.8: Under the Gun: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Administrative (vs Litigated)	-0.024	-0.033	-0.018	-0.020
	(0.022)	(0.020)	(0.019)	(0.023)
Panel B: Direct Effect				
Administrative (vs Litigated)	-0.019	-0.028	-0.022	-0.024
	(0.021)	(0.019)	(0.020)	(0.023)
Log Quantity	-0.006*	-0.005	0.003	0.003
	(0.003)	(0.004)	(0.002)	(0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	63,096	63,096	63,096	63,096
Obs. (Panel B)	63,090	63,090	63,090	63,090
Within R^2 (A)	0.001	0.002	0.000	0.000
Within R^2 (B)	0.002	0.003	0.001	0.001

Notes: Dependent variable: successful tender (LPM). Sample: urgent purchases only (administrative and litigated), all observations, items with both types present. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

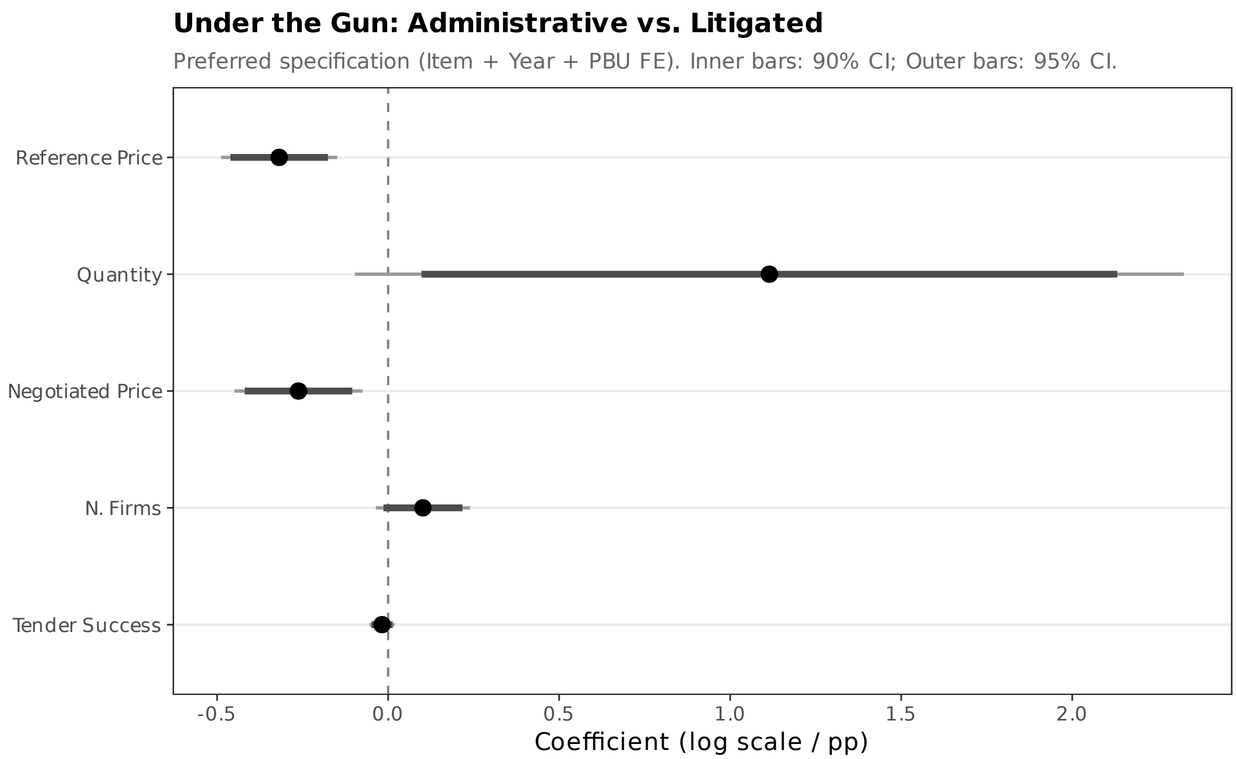

Figure A.1: Under the Gun: Coefficient Estimates with 90% and 95% Confidence Intervals (Preferred Specification)

A.3 Litigated vs. Ordinary: Full Results

Tables A.9–A.13 report the litigated-only estimates. Figure A.2 summarizes the results.

Table A.9: Litigated vs Ordinary: Reference Prices

	(1)	(2)	(3)	(4)
Litigated Purchase	0.259***	0.245***	0.073***	0.075***
	(0.057)	(0.050)	(0.023)	(0.022)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	181,514	181,514	181,511	181,511
Within R^2	0.007	0.007	0.000	0.000

Notes: Dependent variable: log reference price. Sample: litigated and ordinary purchases only (excluding administrative), winners, items with both litigated and ordinary present. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.10: Litigated vs Ordinary: Quantities

	(1)	(2)	(3)	(4)
Litigated Purchase	-0.527***	-0.538***	-0.232***	-0.241***
	(0.133)	(0.134)	(0.085)	(0.083)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	181,514	181,514	181,511	181,511
Within R^2	0.011	0.012	0.002	0.002

Notes: Dependent variable: log quantity. Sample: litigated and ordinary purchases only (excluding administrative), winners. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.11: Litigated vs Ordinary: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Litigated Purchase	0.227***	0.210***	0.087***	0.091***
	(0.037)	(0.032)	(0.021)	(0.019)
Panel B: Direct Effect				
Litigated Purchase	0.049	0.027	-0.008	-0.008
	(0.035)	(0.031)	(0.028)	(0.027)
Log Quantity	-0.339***	-0.341***	-0.412***	-0.412***
	(0.026)	(0.026)	(0.025)	(0.025)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	181,502	181,502	181,499	181,499
Within R^2 (A)	0.006	0.005	0.001	0.001
Within R^2 (B)	0.325	0.337	0.370	0.370

Notes: Dependent variable: log negotiated price. Sample: litigated and ordinary purchases only (excluding administrative), winners. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.12: Litigated vs Ordinary: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Litigated Purchase	-0.116***	-0.136***	-0.082***	-0.081***
	(0.023)	(0.020)	(0.022)	(0.023)
Panel B: Direct Effect				
Litigated Purchase	-0.076***	-0.095***	-0.052***	-0.051***
	(0.017)	(0.014)	(0.016)	(0.017)
Log Quantity	0.063***	0.065***	0.066***	0.066***
	(0.004)	(0.004)	(0.003)	(0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	76,942	76,942	76,938	76,938
Obs. (Panel B)	76,930	76,930	76,926	76,926
Within R^2 (A)	0.005	0.007	0.001	0.001
Within R^2 (B)	0.060	0.067	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: litigated and ordinary purchases only (excluding administrative), winners. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.13: Litigated vs Ordinary: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Litigated Purchase	0.035***	0.037***	0.024***	0.024***
	(0.010)	(0.010)	(0.009)	(0.009)
Panel B: Direct Effect				
Litigated Purchase	0.038***	0.040***	0.027***	0.028***
	(0.010)	(0.010)	(0.009)	(0.009)
Log Quantity	0.006	0.006	0.014***	0.014***
	(0.004)	(0.004)	(0.002)	(0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	208,613	208,613	208,609	208,609
Obs. (Panel B)	208,590	208,590	208,586	208,586
Within R^2 (A)	0.001	0.002	0.000	0.000
Within R^2 (B)	0.002	0.003	0.005	0.005

Notes: Dependent variable: successful tender (LPM). Sample: litigated and ordinary purchases only (excluding administrative), all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

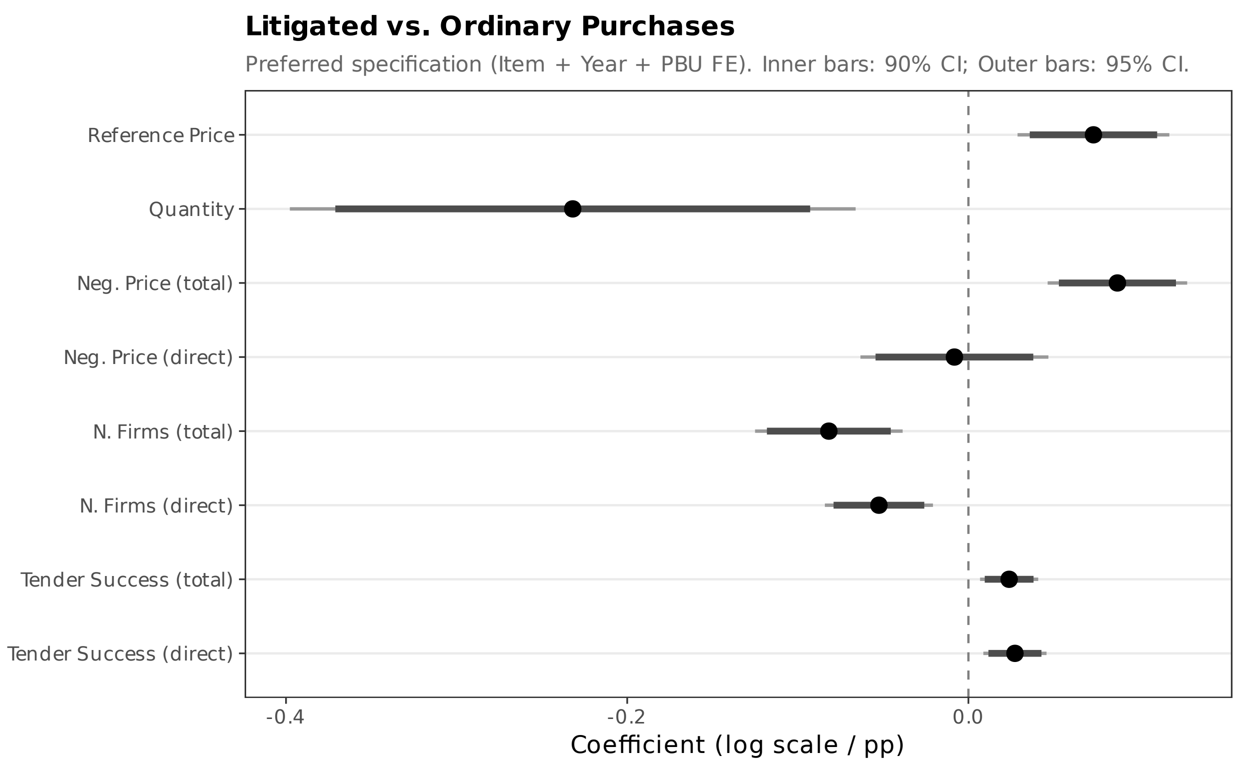

Figure A.2: Litigated vs. Ordinary: Coefficient Estimates with 90% and 95% Confidence Intervals (Preferred Specification)

A.4 Administrative vs. Ordinary: Full Results

Tables A.14–A.18 report estimates comparing administrative and ordinary purchases. Figure A.3 summarizes the results.

Table A.14: Administrative vs. Ordinary: Reference Prices

	(1)	(2)	(3)	(4)
Administrative Purchase	-0.110	-0.104	-0.095*	-0.093*
	(0.070)	(0.073)	(0.049)	(0.047)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	168,752	168,752	168,750	168,750
Within R^2	0.001	0.001	0.000	0.000

Notes: Standard errors clustered at the PBU level in parentheses. Sample restricted to items with both administrative and ordinary purchases. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.15: Administrative vs. Ordinary: Quantities

	(1)	(2)	(3)	(4)
Administrative Purchase	0.514	0.503	0.457	0.448
	(0.481)	(0.474)	(0.331)	(0.327)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	168,752	168,752	168,750	168,750
Within R^2	0.005	0.005	0.004	0.003

Notes: Standard errors clustered at the PBU level in parentheses. Sample restricted to items with both administrative and ordinary purchases. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.16: Administrative vs. Ordinary: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Administrative Purchase	-0.040	-0.030	-0.062	-0.060
	(0.067)	(0.070)	(0.050)	(0.050)
Panel B: Direct Effect (controlling for quantity)				
Administrative Purchase	0.108	0.116	0.109	0.108
	(0.086)	(0.081)	(0.089)	(0.088)
Log Quantity	-0.288***	-0.290***	-0.374***	-0.374***
	(0.024)	(0.023)	(0.027)	(0.027)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	168,739	168,739	168,737	168,737
Within R^2 (A)	0.000	0.000	0.000	0.000
Within R^2 (B)	0.290	0.305	0.351	0.351

Notes: Standard errors clustered at the PBU level in parentheses. Sample restricted to items with both administrative and ordinary purchases. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.17: Administrative vs. Ordinary: Bidder Participation

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Administrative Purchase	-0.066	-0.055	-0.027	-0.023
	(0.051)	(0.048)	(0.029)	(0.028)
Panel B: Direct Effect (controlling for quantity)				
Administrative Purchase	-0.129***	-0.117***	-0.037*	-0.033
	(0.018)	(0.014)	(0.020)	(0.020)
Log Quantity	0.067***	0.069***	0.071***	0.071***
	(0.005)	(0.005)	(0.005)	(0.005)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,808	81,808	81,808	81,808
Obs. (Panel B)	81,799	81,799	81,799	81,799
Within R^2 (A)	0.001	0.001	0.000	0.000
Within R^2 (B)	0.066	0.071	0.046	0.047

Notes: Standard errors clustered at the PBU level in parentheses. Sample restricted to items with both administrative and ordinary purchases. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.18: Administrative vs. Ordinary: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Administrative Purchase	-0.007	-0.013	0.009	0.009
	(0.021)	(0.020)	(0.011)	(0.011)
Panel B: Direct Effect (controlling for quantity)				
Administrative Purchase	-0.010	-0.015	0.001	0.001
	(0.021)	(0.021)	(0.013)	(0.013)
Log Quantity	0.004	0.004	0.015***	0.015***
	(0.005)	(0.005)	(0.002)	(0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	196,988	196,988	196,986	196,986
Obs. (Panel B)	196,971	196,971	196,969	196,969
Within R^2 (A)	0.000	0.000	0.000	0.000
Within R^2 (B)	0.000	0.001	0.005	0.005

Notes: Standard errors clustered at the PBU level in parentheses. Sample restricted to items with both administrative and ordinary purchases. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

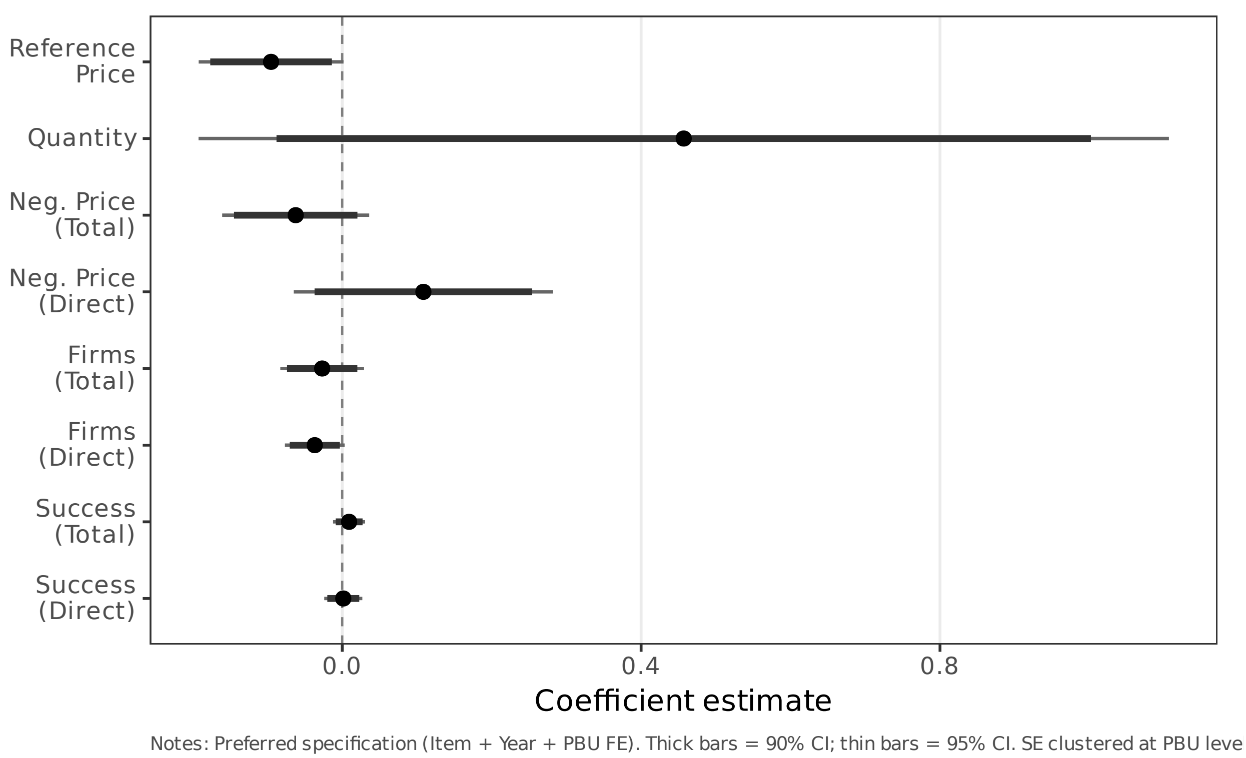

Figure A.3: Administrative vs. Ordinary: Coefficient Estimates with 90% and 95% Confidence Intervals (Preferred Specification)

A.5 Heterogeneity

Tables A.19–A.22 report heterogeneity analysis along four dimensions.

Table A.19: Heterogeneity: SUS Component

	Split Sample		Interaction
	Specialized	Basic SUS	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.005	0.030*	-0.073
	(0.046)	(0.016)	(0.097)
Urgent × Basic SUS			0.159
			(0.115)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	45,048	151,835	196,883
Within R^2	0.000	0.000	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.20: Heterogeneity: Time Period

	Split Sample		Interaction
	Early	Late (2014+)	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.074***	0.045**	0.259***
	(0.025)	(0.018)	(0.035)
Urgent × Late Period			-0.339***
			(0.043)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	82,658	113,936	196,883
Within R^2	0.000	0.000	0.005

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.21: Heterogeneity: Market Competition

	Split Sample		Interaction
	Low Comp.	High Comp.	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.046***	0.143***	0.026
	(0.016)	(0.034)	(0.018)
Urgent × High Competition			0.226***
			(0.044)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	125,950	61,313	187,264
Within R^2	0.000	0.001	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

Table A.22: Heterogeneity: PBU Size

	Split Sample		Interaction
	Small PBU	Large PBU	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.023	0.062***	-0.000
	(0.026)	(0.018)	(0.049)
Urgent × Large PBU			0.070
			(0.054)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	31,902	164,511	196,883
Within R^2	0.000	0.000	0.000

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

A.6 Winsorization Sensitivity

Tables A.23–A.26 replicate the main results under three winsorization levels: none (Panel A), 1%/99% baseline (Panel B), and 5%/95% (Panel C). Results are robust across all levels.

Table A.23: Robustness: Reference Prices

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.167***	0.159***	0.026*	0.026*
	(0.056)	(0.051)	(0.014)	(0.014)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.164***	0.156***	0.027*	0.027*
	(0.056)	(0.050)	(0.014)	(0.014)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.156***	0.150***	0.027*	0.028**
	(0.053)	(0.048)	(0.014)	(0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896

Notes: Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.24: Robustness: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.160***	0.152***	0.051***	0.054***
	(0.043)	(0.038)	(0.015)	(0.016)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.162***	0.153***	0.058***	0.061***
	(0.041)	(0.037)	(0.017)	(0.017)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883

Notes: Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.25: Robustness: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.015)	(0.015)
Panel B: Winsorized 1%/99%				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.014)	(0.015)
Panel C: Winsorized 5%/95%				
Urgent Purchase	-0.100***	-0.111***	-0.055***	-0.053***
	(0.022)	(0.021)	(0.014)	(0.015)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (A)	81,377	81,377	81,373	81,373
Obs. (B)	81,805	81,805	81,801	81,801
Obs. (C)	81,805	81,805	81,801	81,801

Notes: Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.26: Robustness: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	226,305	226,305	226,301	226,301

Notes: Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

A.7 Under the Gun: Progressive Controls

Table A.27 examines the sensitivity of the UTG estimate to progressive inclusion of control variables.

Table A.27: Robustness: Under the Gun with Progressive Controls

	(1)	(2)	(3)	(4)	(5)
Panel A: No Winsorization					
Administrative (vs Litigated)	-0.259**	0.138	0.115**	0.201**	0.195**
	(0.095)	(0.130)	(0.054)	(0.090)	(0.087)
Panel B: Winsorized 1%/99%					
Administrative (vs Litigated)	-0.262**	0.117	0.110*	0.199**	0.194**
	(0.096)	(0.122)	(0.055)	(0.093)	(0.090)
Panel C: Winsorized 5%/95%					
Administrative (vs Litigated)	-0.261**	0.060	0.089*	0.169*	0.165*
	(0.096)	(0.093)	(0.049)	(0.084)	(0.080)
Log Quantity	No	Yes	Yes	Yes	Yes
Log Ref. Price	No	No	Yes	Yes	Yes
Log N. Firms	No	No	No	Yes	Yes
Item FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Year-Month FE	No	No	No	No	Yes
PBU FE	Yes	Yes	Yes	Yes	Yes
Obs. (A)	56,803	56,803	56,803	14,913	14,913
Obs. (B)	56,803	56,803	56,803	15,052	15,052
Obs. (C)	56,803	56,803	56,803	15,052	15,052

Notes: Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

A.8 Geographic Distribution

Figures A.4–A.6 present per capita choropleth maps for each purchase type.

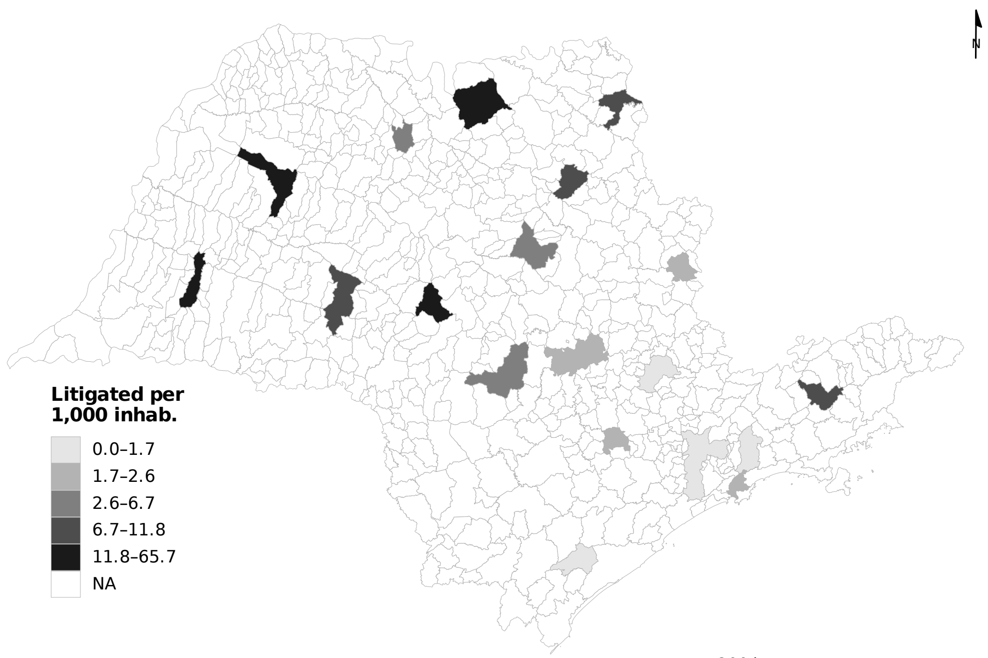

Figure A.4: Litigated Purchases per 1,000 Inhabitants (São Paulo, 2009–2019)

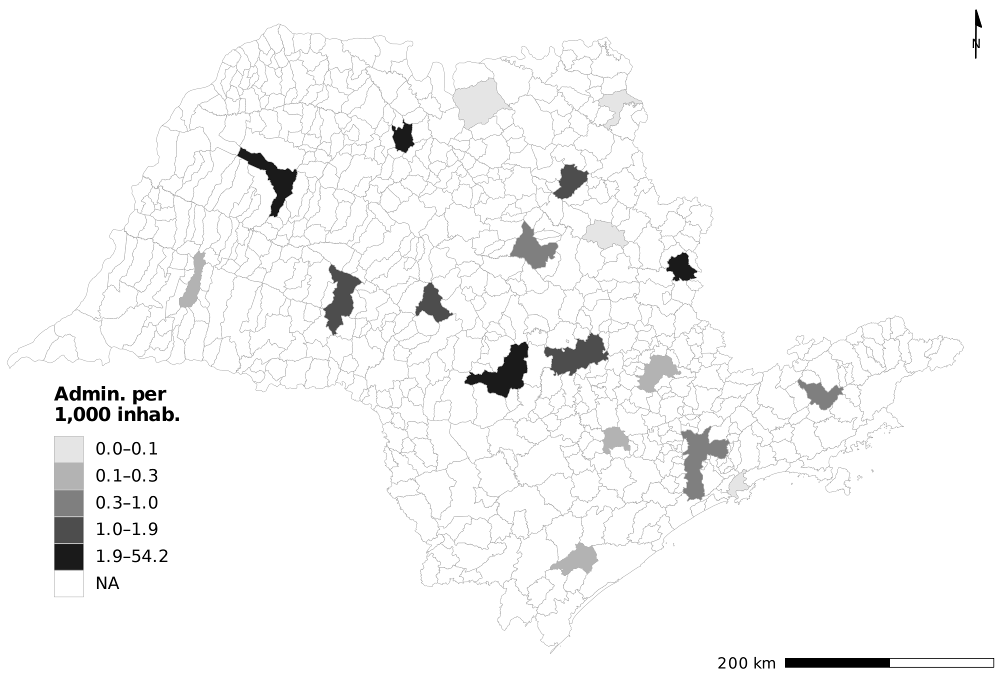

Figure A.5: Administrative Purchases per 1,000 Inhabitants (São Paulo, 2009–2019)

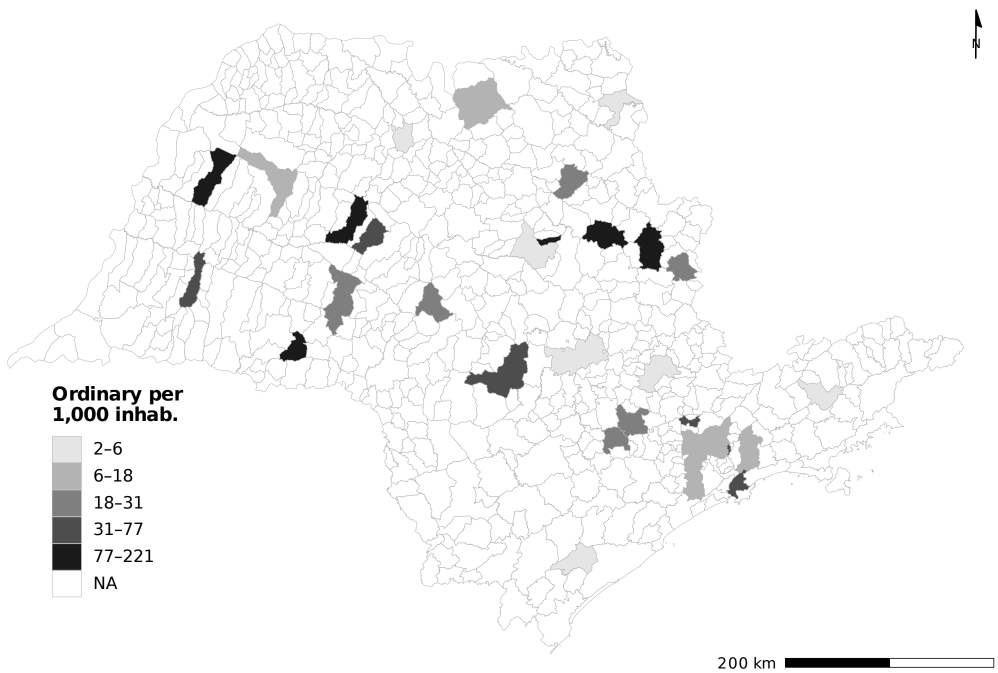

Figure A.6: Ordinary Purchases per 1,000 Inhabitants (São Paulo, 2009–2019)

A.9 Distributional Evidence

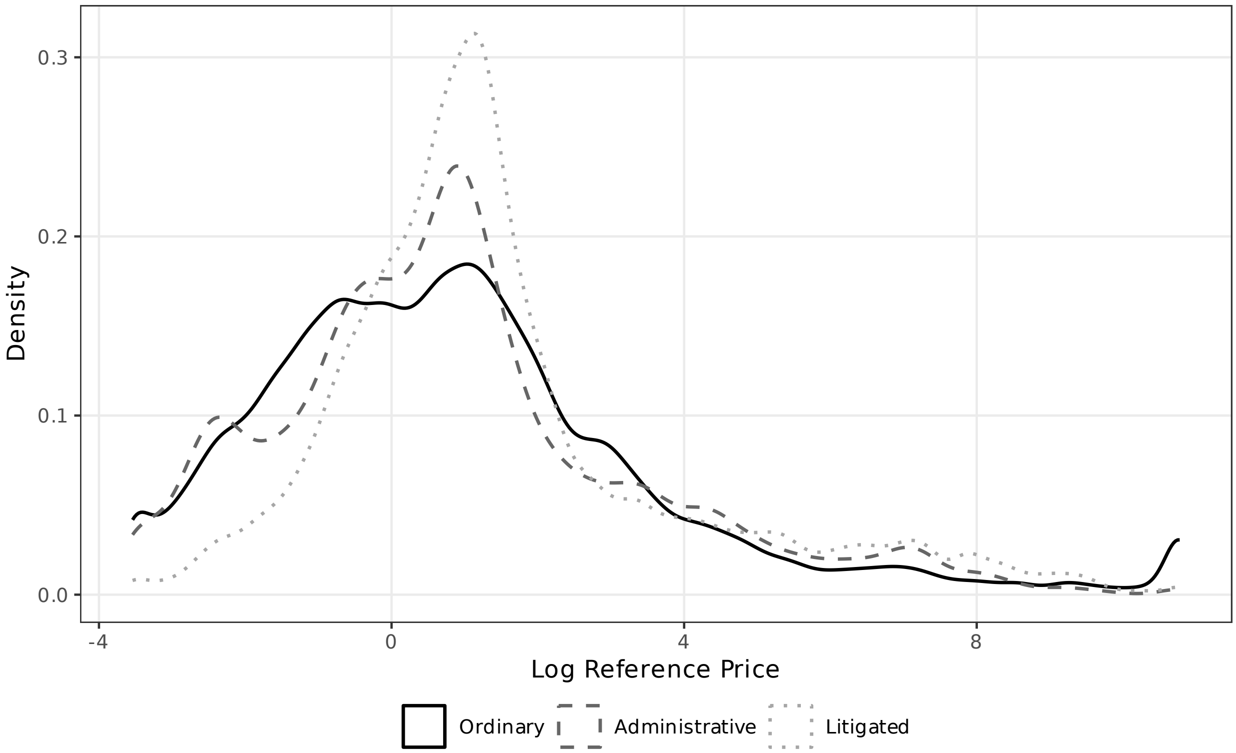

Figure A.7: Distribution of Log Reference Price by Purchase Type

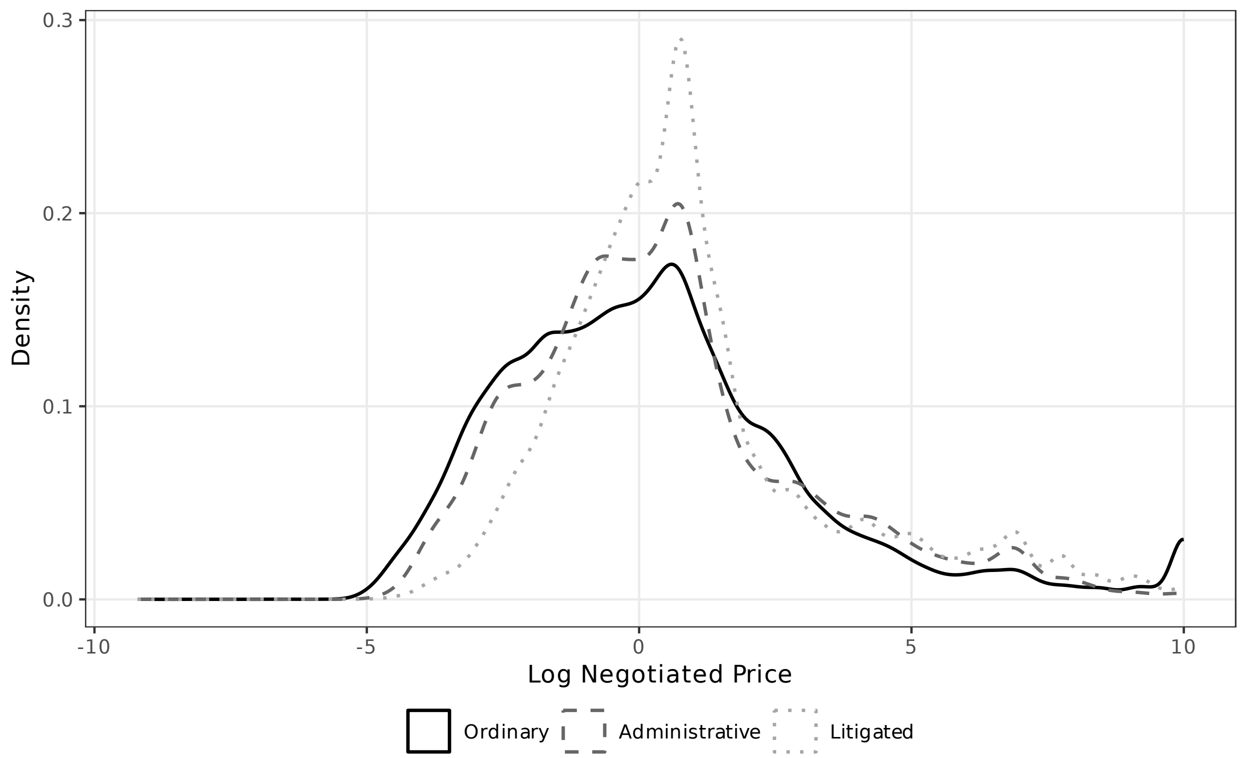

Figure A.8: Distribution of Log Negotiated Price by Purchase Type

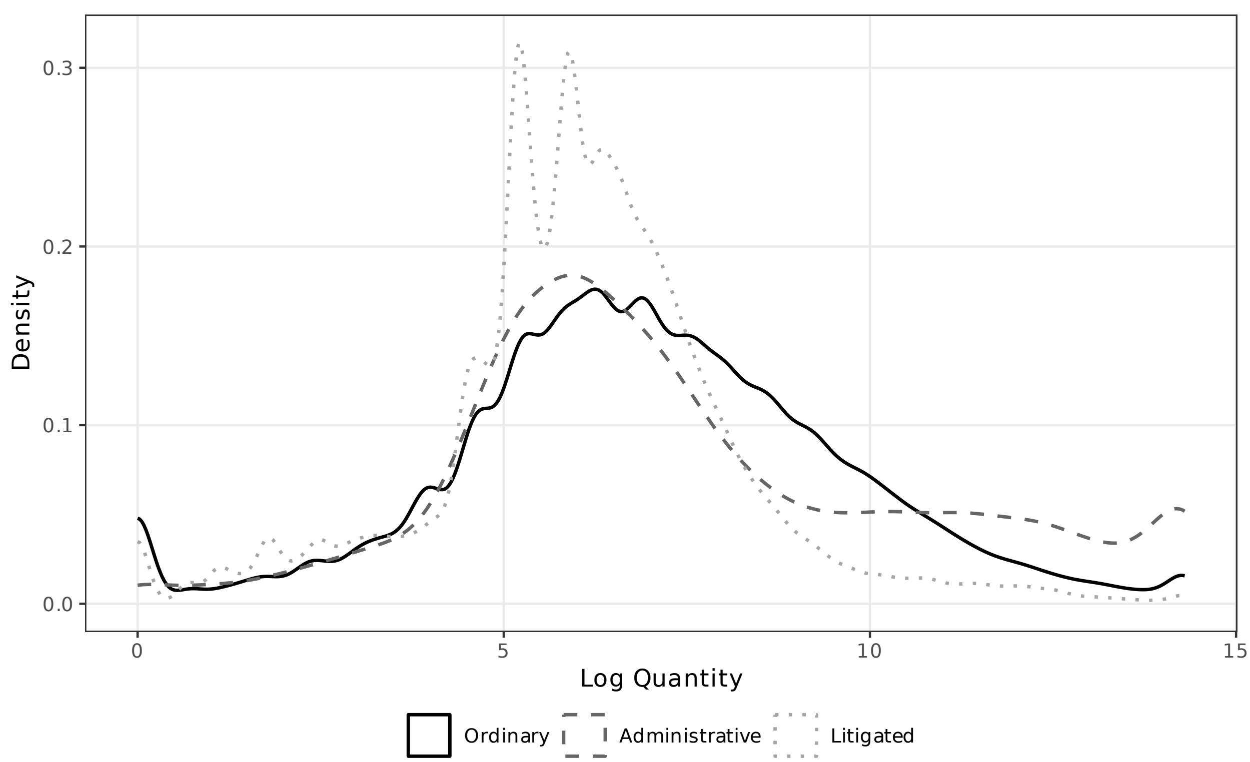

Figure A.9: Distribution of Log Quantity by Purchase Type

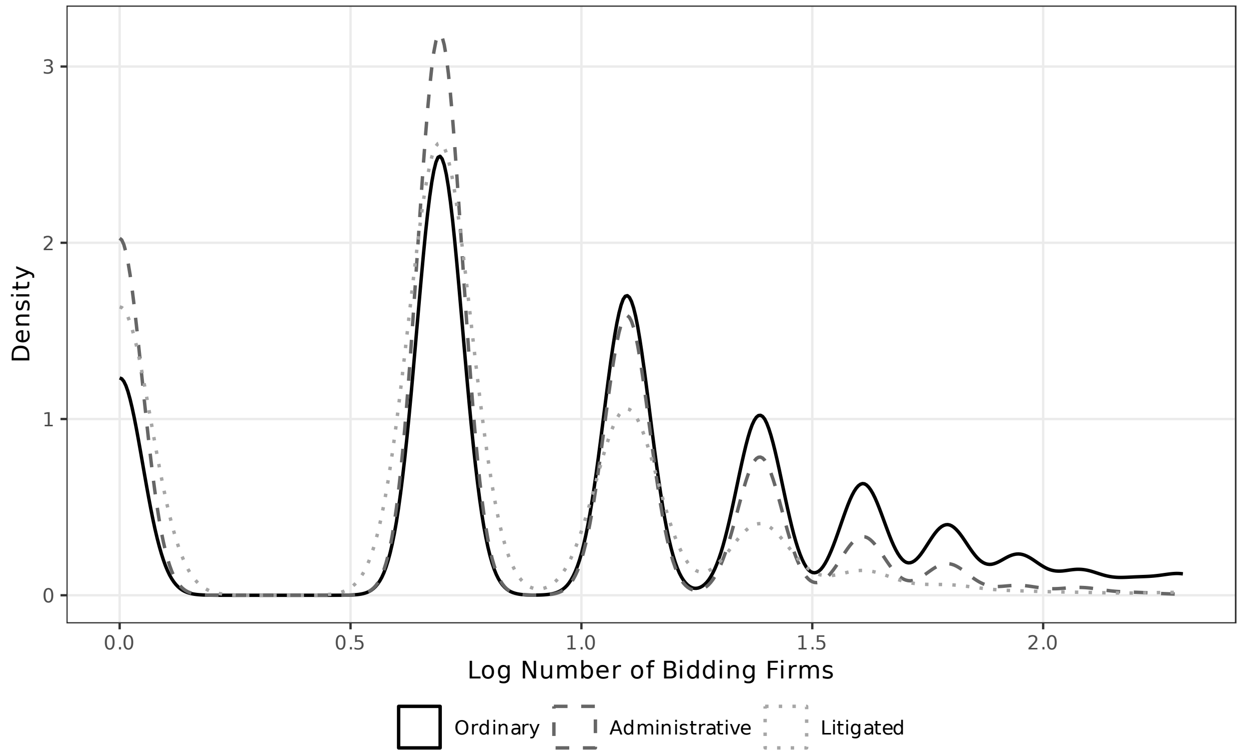

Figure A.10: Distribution of Log Number of Bidding Firms by Purchase Type

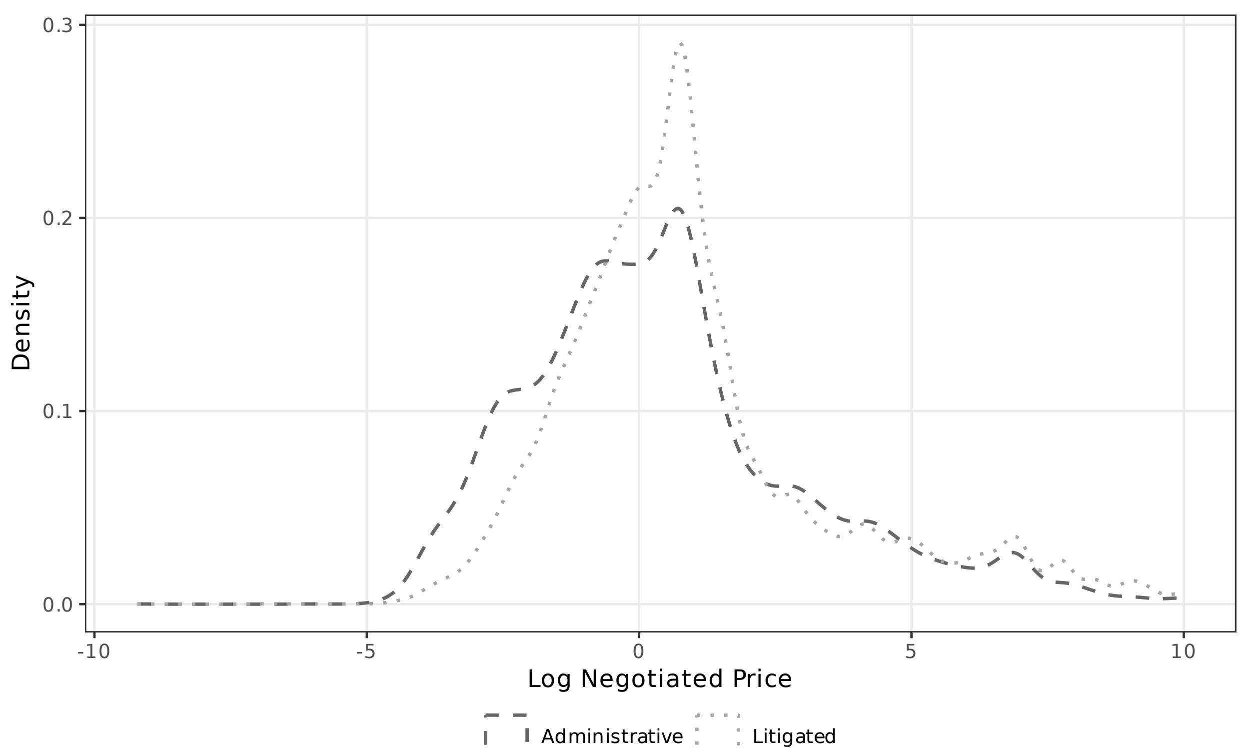

Figure A.11: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

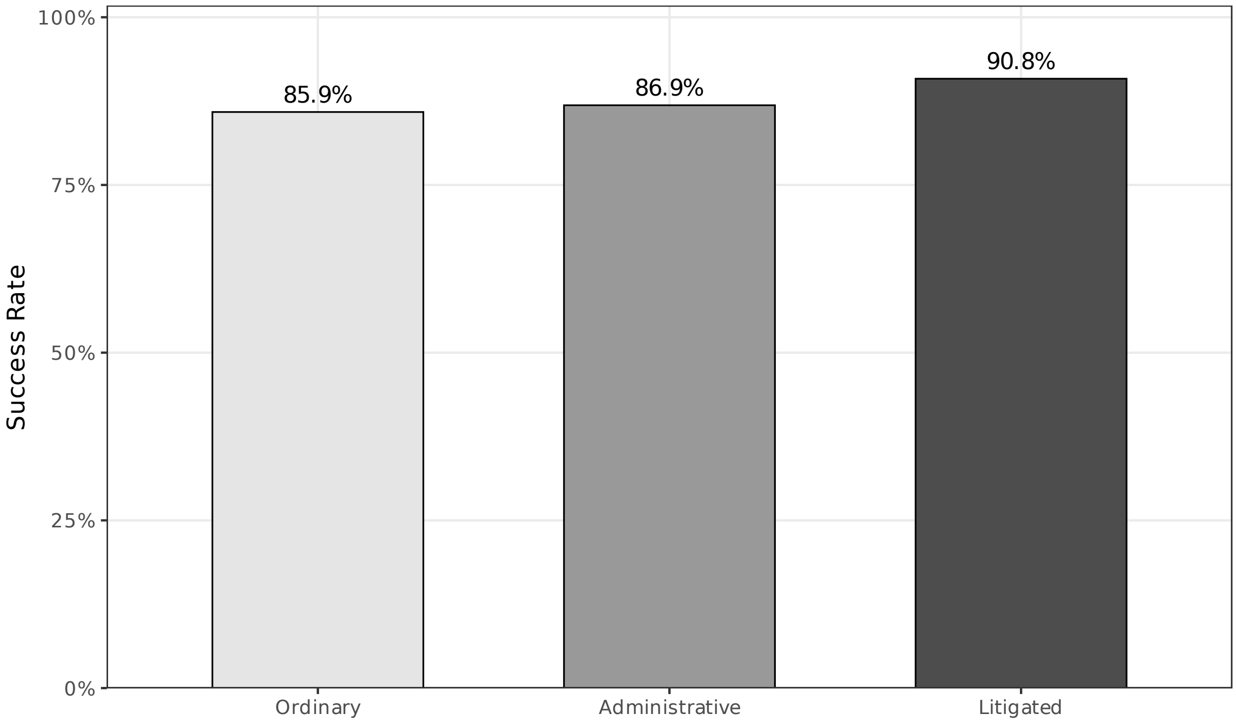

Figure A.12: Tender Success Rate by Purchase Type

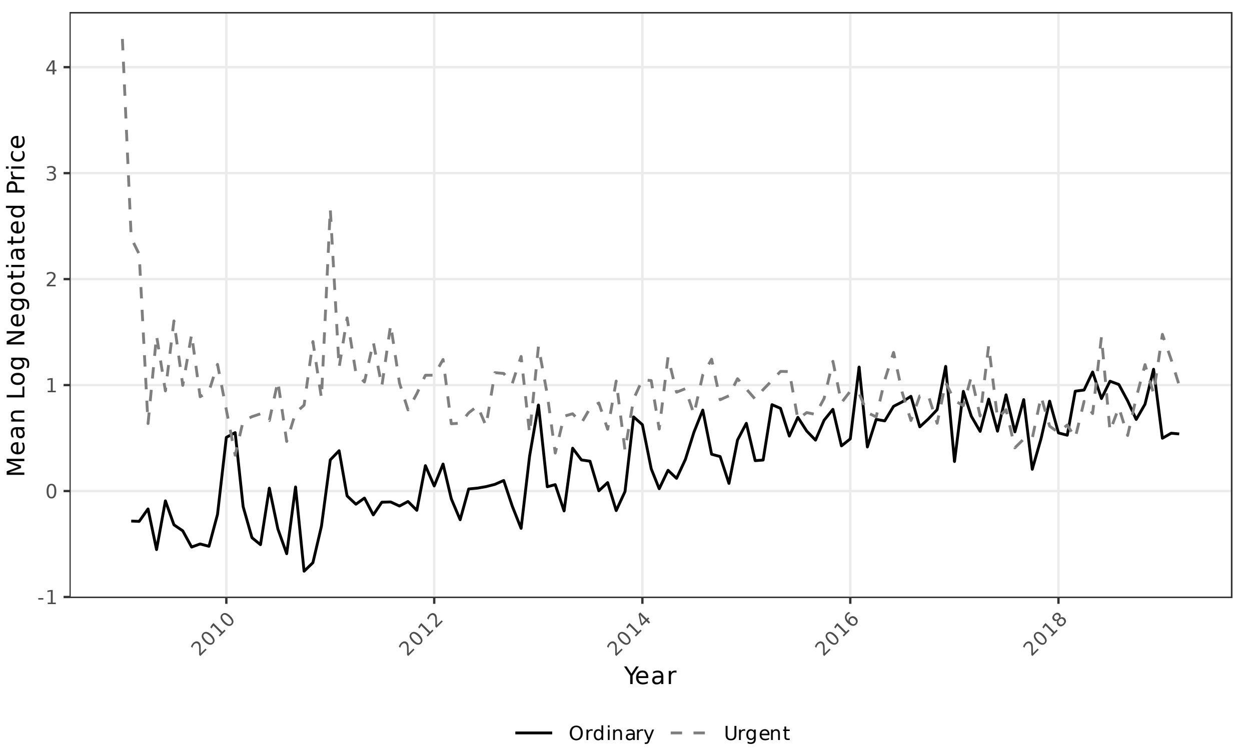

Figure A.13: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent